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Editor-in-Chief
Computers, Environment and Urban Systems

Dear Editor,

I am pleased to submit our manuscript entitled *"Quantifying the accuracy of OpenStreetMap for longitudinal cycling-infrastructure change: A reproducible validation framework using Google Street View (Barcelona, 2015–2023)"* for consideration in the Open Urban Data Science special issue of *Computers, Environment and Urban Systems*.

This study addresses a key challenge in open urban data science: the absence of systematically maintained longitudinal data on cycling infrastructure, which constrains robust evaluation of network expansion and its impacts on mobility, safety, and equity. We develop and implement a fully reproducible open-source computational framework that reconstructs historical cycling networks from OpenStreetMap (OSM) extracts, performs geometric snapshot differencing, and quantifies detection accuracy using probability-based stratified validation against Google Street View imagery.

Beyond empirical findings, the core contribution of the paper is methodological and computational. The workflow provides an open and transferable implementation for detecting and calibrating infrastructure change using volunteered geographic information. By explicitly quantifying precision and recall, and by documenting each step of the pipeline, the framework supports more reliable longitudinal inference from open urban data. The full computational workflow is implemented in R and is publicly available in an open-source repository (MIT licence) at: https://github.com/GEMOTT/osm-cycling-change-framework. The repository contains all scripts, documentation, and supplementary materials required to reproduce the analyses and regenerate the manuscript outputs. A permanent archival DOI (e.g. via Zenodo) will be created upon acceptance.

Our results show that OSM snapshot differencing reliably captures infrastructure additions but substantially over-detects removals due to representational edits. By proposing calibration strategies based on validated detection metrics, the framework strengthens exposure measurement in transport, public health, and urban equity research.

We believe this work aligns closely with the aims of the Open Urban Data Science special issue, as it advances transparent, reproducible, and open computational methods for urban infrastructure analysis while critically assessing the reliability of widely used open datasets.

We confirm that this manuscript has not been published previously, is not under consideration elsewhere, and that all authors have approved the manuscript and agree with its submission.

Thank you for considering our manuscript. We look forward to your response.

Sincerely,

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